Question Paper

Artificial Intelligence

No. of Questions: 50, Duration: 60Min.

1. Which of the following is NOT a benefit of using cross-validation in model evaluation?

A. It provides a more reliable estimate of model performance on unseen data.

B. It helps in selecting hyperparameters by comparing different models.

C. It completely eliminates the possibility of overfitting.

D. It maximizes the use of available data for both training and validation.

2. A binary classifier has an accuracy of 70% on a test set. The confusion matrix is as follows:

	Predicted Positive	Predicted Negative
Actual Positive	**60**	**40**
Actual Negative	**20**	**80**

What is the precision of the classifier for the positive class?

A. 60% B. 75% C. 66.7% D. 50%

3. Which of the following techniques is unique to CatBoost compared to other gradient boosting algorithms?

A. Using random subsampling of data at each boosting step
B. Implementing L2 regularization for leaf-wise splits
C. Ordered boosting to avoid target leakage in categorical feature handling
D. Utilizing histogram-based learning to speed up training

4. A dataset has 95% of instances belonging to class A and 5% to class B. A classifier predicts all instances as class A, achieving 95% accuracy. Which metric should you use to better evaluate the model, and what would be its value for class B?

A. Precision; Undefined for class B. B. Recall; 0% for class B.
C. F1-Score; 0% for class B. D. Accuracy is sufficient; the model performs well.

5. In XGBoost, what is the role of the gradient in the objective function?

A. Scaling optimal feature importance B. Avoid overfitting

C. Improve convergence D.Controls the model's depth

---

6. What is the purpose of regularization in machine learning?

A. To increase model complexity B. To reduce overfitting
C. To improve training speed D. To normalize data

---

7. Which of the following is a characteristic of ensemble learning?

A. Uses a single model for predictions B. Combines multiple models to improve accuracy
C. Requires labeled data only D. Is limited to linear algorithms

---

8. Which of the following activation function has a range of values from 0 to 1?

A. Sigmoid B. tanh C. ReLU D. None of the Above

---

9. Which of the following is NOT true about transfer learning?

A. Transfer learning requires re-training the entire model from scratch.

B. Transfer learning allows the use of pre-trained models on new, similar tasks.
C. Transfer learning can sometimes be effective with limited training data.
D. Transfer learning can be used to accelerate model convergence.

---

10. A fully connected neural network (FNN) with two hidden layers, each having 5 neurons. If the input layer has 10 neurons and the output layer has 1 neuron, how many total parameters FNN has?

A. 80 B. 105 C. 60 D. 120

---

11. Given a dataset with 1,000 samples and 10 features, you want to use a batch size of 32 and an epoch size of 50. How many iterations will the training take per epoch?

A. 31 B. 30 C. 32 D. 28

---

12. Which layer in a convolutional neural network can help effectively reduce the spatial dimensions of a feature map thereby reducing the computation cost?

A. Convolution layer B. Fully connected layer

C. Activation function layer D. Pooling layer

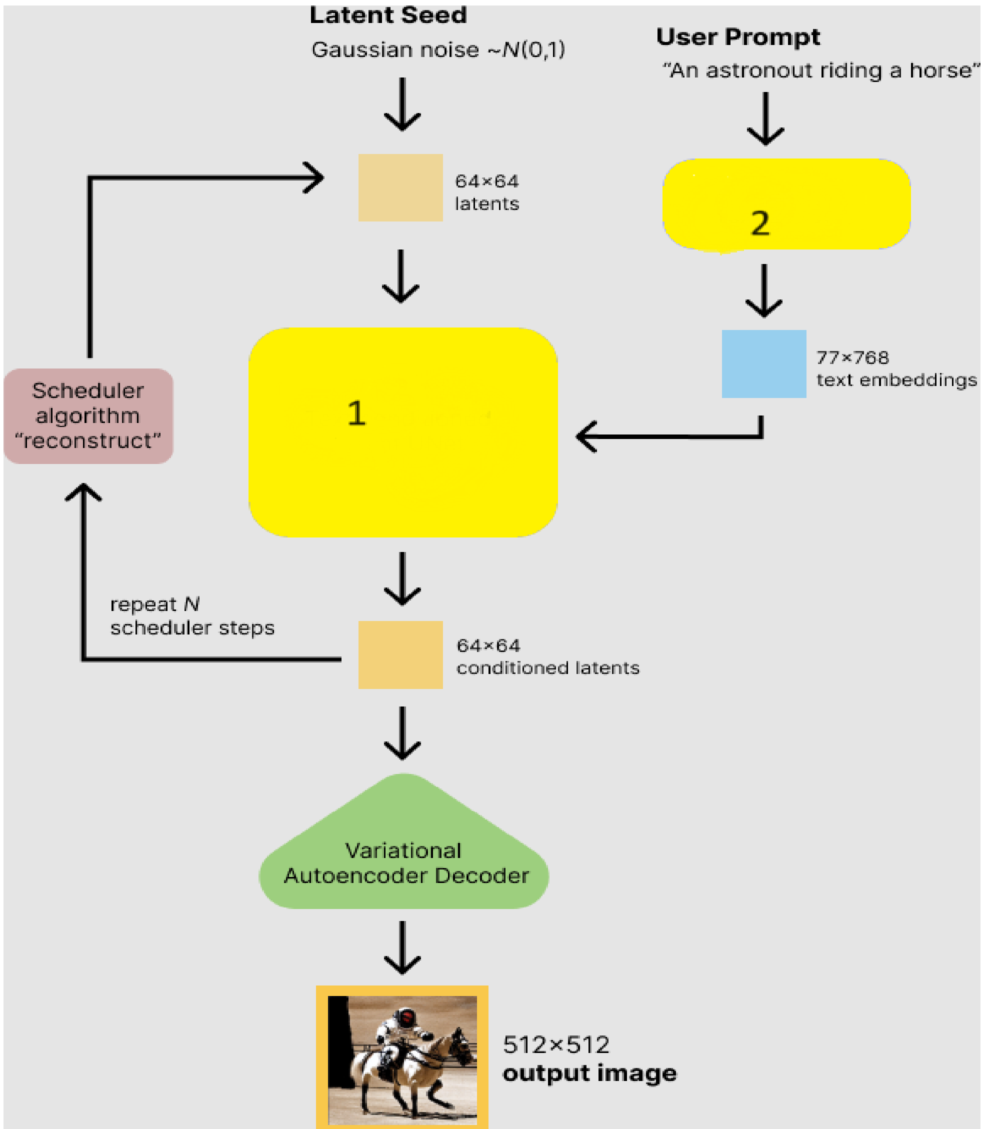
---

13. Which of the following method provides dynamic embeddings?

A. Word2Vec B. FastText C. GloVe D. BERT

---

14. Given the Diffusion Model architecture, what are the names of components 1 and 2 (represented using yellow boxes)?



A. Text Condition latent Unet, Frozen Clip text encoder

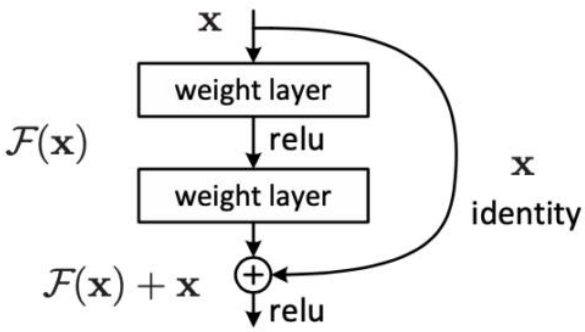
B. Conditional encoder latent Unet, Frozen Clip text decoder

C. Latent Text encoder with Unet , Frozen Clip noise decoder

D. Unet latent encoder, Frozen Clip noise encoder

---

15. The ResNet architecture introduces the simple concept of adding an intermediate input to the output of a series of convolution blocks. The building block of a residual network is given below. Which of the following statements is FALSE?



A. Skip connections will not add additional computational cost during training the network

B. Skip connections will increase the impact of vanishing gradient problem

C. ResNet stacks multiple identity mappings, skips those layers

D. None of the Above

---

16. What is the full form of BERT in Natural Language Processing?

A. Bi-directional Encoder Representations from Transformers

B. Bi-directional Embedding Representations from Transformers

C. Binary Encoder Representations from Transformers

D. Balanced Embedding Representations from Transformers

---

17. What is the primary function of a discriminator in a Generative Adversarial Network (GAN)?

A. To generate realistic images
B. To minimize the loss of the generator
C. To classify real vs fake samples
D. To reduce computational complexity

---

18. What is the key advantage of subword tokenization methods like BPE or WordPiece in NLP models?

A. They can handle out-of-vocabulary (OOV) words effectively

B. They split every word into individual characters

C. They ignore rare words to save memory

D. They always maintain the original form of every word

---

19. In Transformer models for text generation, what is the main role of attention mechanisms?

A. They reduce model complexity by limiting input length.
B. Allow model to focus on different parts of the input for context-aware generation.
C. They focus only on the output sequence, ignoring the input.
D. They enforce a fixed alignment between input and output tokens.

---

20. What is a key advantage of self-attention in Transformers over RNNs for long text generation?

A. It processes input sequentially for coherence.
B. It considers all tokens simultaneously, capturing long-range dependencies.
C. It relies only on the last hidden state for speed.
D. It focuses only on nearby tokens.

21. Which model architecture is commonly used to transform 2D images into 3D scenes?

A. AlphaZero B. Neural Radiance Fields (NeRF) C. GPT-4 D. Stable Diffusion

---

22. Which of the following tasks commonly use cross-attention mechanisms?

A. Word embedding generation B. Text classification

C. Machine translation D. Sentiment analysis

---

23. Which mathematical function is commonly used to generate positional encodings in transformers?

A. Polynomial functions B. Sine and cosine functions

C. Exponential functions D. Linear functions

---

24. What does the acronym RAG stand for in the context of natural language processing?

A. Retrieval-Augmented Generation B. Randomized Action Generation

C. Reinforced Adversarial Generation D. Recursive Algorithm Generation

---

25. In cross-attention, what is the role of the query (Q) from one sequence?
A. To match with keys from the same sequence

B. To compute attention weights over another sequence's keys
C. To normalize the attention output
D. To apply softmax on input tokens

---

26. Consider the following Python code snippet. What is the output of the print statement, and what concept is being demonstrated?

```
a = [1, 2, 3]
b = a[:]
a[0] = 5
print(b[0])
```

A. 1, demonstrates shallow copy
B. 5, demonstrates shallow copy
C. 1, demonstrates deep copy
D. 5, demonstrates deep copy

---

27. What is the return value of the following recursive Python?

```
def foo(x):
      if x <= 1:
            return 1
      return x * foo(x - 1)

result = foo(5)
print(result)
```

A. 120
B. 25
C. Error
D. 5

---

28. What is the output of the following Python code snippet?

```
import threading
def print_numbers():
      for i in range(3):
            print(i)

t = threading.Thread(target=print_numbers)
t.start()
t.join()
```

A. `0 1 2` and terminate normally.
B. `0 1 2`, but the main thread will hang due to `t.join()`.
C. Runtime Error
D. It will print `0 1 2` twice due to multithreading.

---

29. What is the output of the following Python code snippet?

```
def gen():
    yield 1
    yield 2
    yield 3

g = gen()
print(next(g))
print(next(g))
g.close()
print(next(g))
```

A. `1 2`   B. `1 2 3`   C. `1 2 RuntimeError`   D. `1 2 GeneratorExit`

---

30. What will happen if the following code is executed, considering Python's memory management?

```
arr = [1] * (10**7)
del a
```

A. Immediate memory release happens after the execution of del method
B. Memory is not immediately released due to Python's garbage collector
C. Causes an out-of-memory error
D. Raises `MemoryError` due to excessive list size of arr

---

31. What is the output of the following Python code?

```
def print_string(s):
    if len(s) == 0:
        return s
    return print_string(s[1:]) + s[0]

print(print_string("hello"))
```

A. `"olleh"`
B. `"hello"`, as the recursion does nothing
C. `Error` due to exceeding recursion depth
D. `Error` because the base case is incorrect

---

32. What will be the output of the following Python code?

```
f = lambda x: x**2
nums = [1, 2, 3, 4]
result = [f(n) for n in nums if n % 2 == 0]
print(result)
```

A. `[1, 4, 9, 16]` B. `[4, 16]` C. `[2, 4]` D. `Error`

---

33. What does the `strip()` method do in Python?

A. Removes all whitespace from a string B. Removes leading and trailing whitespace

C. Converts a string to lowercase D. Splits a string into a list

---

34. What will be the output of the following code: `print([x**2 for x in range(3)])`?

A. `[0, 1, 2]` B. `[0, 1, 4]` C. `[1, 2, 3]` D. `[1, 4, 9]`

---

35. In PyTorch, what does the `.detach()` method do in the context of a tensor?

A. It copies the tensor into another memory location.
B. It separates the tensor from the computation graph, preventing gradient calculation.
C. It deletes the tensor from memory after use.
D. It will detach a layer of a neural network

---

36. Given a simple neural network in PyTorch with an input tensor of size (64, 128), and a fully connected linear layer that transforms it to (64, 10), what is the number of parameters in the linear layer?

A. 1280 B. 12810 C. 1290 D. 12910

---

37. In a training loop, if your PyTorch model outputs `tensor([1.23, 4.56], requires_grad=True)` and you call `.backward()` without a gradient argument, what happens?

A. Computes the gradient for the output tensor.
B. Raises an error for a missing gradient argument.
C. Computes the gradient with respect to the scalar sum.
D. Computes gradients for each element individually.

---

38. In TensorFlow, given a dropout layer with a dropout rate of 0.5 during training, what happens during inference?

A. Half the neurons are randomly dropped.
B. The neurons remain unchanged, but their outputs are scaled by 0.5.
C. Dropout is disabled and the outputs are scaled by 1/0.5.
D. Dropout is disabled, and no scaling is applied.

---

39. In PyTorch, how can you handle sequences of varying lengths in an RNN?

A. Use `torch.nn.RNN` with `batch_first=True`.
B. Use `torch.nn.RNN` and manually pad sequences.
C. Use `torch.nn.utils.rnn.pack_padded_sequence` before `torch.nn.RNN`.
D. Use `torch.nn.RNN` with `torch.no_grad()`.

---

40. Which component of AIOps is responsible for identifying potential issues before they occur, based on historical patterns?

A. Incident management
B. Anomaly detection
C. Event correlation
D. Root cause analysis

---

41. In an AIOps workflow, which stage focuses on learning from past incidents to improve future responses?

A. Incident management
B. Continuous feedback loop
C. Event correlation
D. Predictive maintenance

---

42. During the lifecycle of Large Language Models (LLMs) development, which of the following steps is NOT directly involved in the training process?

A. Feature Engineering
B. Pre-training
C. Fine-tuning
D. Reinforcement Learning from Human Feedback (RLHF)

---

43. The following are the notations used in error decomposition in machine learning. Which of the following statements are TRUE?

- **H - Hypothesis Space**
- **I - Number of train data samples**
- **P- Tolerance of the optimizer**

A. If H increases, I and P are fixed then the approximation error decreases
B. If H increases, I and P are fixed then the estimation error decreases

C. If H and P are fixed, I increases then the estimation error decreases
D. If H and I are fixed, P increases then the optimization error decreases

---

44. Which of the following sequences represents the most likely order of training stages for Large Language Models (LLMs)?

i. Pre-training
ii. Reinforcement Learning from Human Feedback (RLHF)
iii. Instruction Fine-tuning

A. i → iii → ii B. ii → i → iii C. iii → i → ii D. i → ii → iii

---

45. When both top-k and top-p sampling are set together in token selection, what is the correct order of execution?

A. Top-k acts before top-p B. Top-p acts before top-k
C. Top-k has no effect D. Top-p has no effect

---

46. What are the key steps involved in the Retrieval-Augmented Generation (RAG) pipeline?

A. Retrieval, Generation, Ranking B. Ranking, Generation, Retrieval
C. Retrieval, Ranking, Generation D. Generation, Retrieval, Ranking

---

47. Which of the following metrics is NOT typically used for evaluating the quality of factual language summaries generated by a Large Language Model (LLM)?

A. ROUGE Score B. BLEU Score
C. Perplexity D. F1 Score

---

48. What is the term for the technique that utilizes a smaller model to learn from a larger pre-trained model, thereby improving efficiency?

A. Meta Learning B. Transfer Learning
C. Knowledge Distillation D. Few Shot Learning

---

49. Which of the following techniques is used to prevent the overfitting in convolutional neural networks?

A. Early stopping B. Data augmentation
C. Batch normalization D. All of the above

---

50. Which of the following machine learning algorithms is a non-parametric lazy classifier?

A. Single Perceptron B. KNN C. XGBoost D. SVM